To avoid wasting net bandwidth and server computations, usually the requests to autocomplete are sent every few seconds or when a user stops pausing. When a response comes back, the page updates the list of entries shown.
This is still not ideal, as with HTTP/1.1 it is not possible to cancel requests that have already been sent and reached the server. Moreover, we have no guarantee on the order in which responses come back. Therefore, if more characters are typed or erased before the previous response comes back, we'll have to somehow keep some kind of versioning for the responses and never replace the results shown if a stale response arrives.
This will be mitigated with HTTP/2, since it introduces cancellable requests, among other cool things.

6.5 Summary

- **Handling strings**: there is a fundamental difference between working with primitive data types like integers or floats, and working with strings: while all integers will require the same amount of memory to be stored[20], strings can be of arbitrary length, and hence require arbitrarily many bytes.
- **Optimizing performance:** it follows that the length of the string on which a data structure works is an important factor in their asymptotic analysis; this leaves space for further optimizations that are not possible with the simplest data types.
- **Trie:** this data structure allows to more efficiently store and query large sets of strings, assuming many of them share some common prefixes.